Multi-Agent Reinforcement Learning for Autonomous Urban Traffic Flow Optimization

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Abstract - Urban traffic congestion imposes severe and growing economic, environmental, and social costs on cities worldwide. Traditional signal control methods fixed-time plans and rule-based adaptive systems rely on manually calibrated parameters that fail to respond to the dynamic and heterogeneous nature of real-world traffic demand. This paper presents a decentralized Multi-Agent Reinforcement Learning (MARL) framework for autonomous traffic signal control using independent Deep Q-Network (DQN) agents, deployed on the real-world Cologne8 benchmark corridor. Each of the eight agents independently observes local lane-level queue states and selects optimal signal phase transitions without inter-agent communication. A custom hybrid reward function simultaneously penalizes total halting vehicles and the variance of per-lane queue lengths, promoting both throughput and load equity. The system is trained for 650,000 timesteps via a two-phase protocol with progressive learning rate decay. Empirical evaluation against a fixed-time baseline yields a 95.1% reduction in average cumulative waiting time (285.95s to 13.88s per step), an 81.1% reduction in maximum wait spike, a 79.7% reduction in average stopped cars, and a 72.4% reduction in maximum gridlock severity. These results demonstrate that the IDQN paradigm with hybrid reward shaping is a highly effective and practically deployable approach to autonomous multi-intersection traffic management.

Index Terms - autonomous traffic control, deep Q-network, multi-agent reinforcement learning, SUMO simulation, urban traffic optimization

I. Introduction

The relentless pace of global urbanization has escalated urban traffic congestion into one of the most pervasive, complex, and economically damaging infrastructure challenges of the 21st century [1]. As municipal populations continue to swell at unprecedented rates, the physical capacity of existing urban road networks is consistently overwhelmed by the sheer volume of daily commuters, commercial logistics, and public transit [2]. The economic toll of these systemic inefficiencies is staggering; recent global mobility scorecards estimate that severe congestion costs major economies tens of billions of dollars annually in lost commuter productivity, disrupted supply chains, and wasted fuel consumption [3]. Furthermore, these massive economic costs are inextricably linked to severe environmental consequences. Stop-and-go traffic, characterized by frequent idling, sudden braking, and harsh accelerations at poorly managed intersections, produces disproportionately high concentrations of greenhouse gases [4]. These emissions include carbon monoxide (CO), nitrogen oxides (NOx), and harmful particulate matter, which severely degrade urban air quality and contribute to public health crises [5]. Consequently, mitigating traffic congestion is no longer just an issue of logistical convenience, but a critical, foundational imperative for the sustainable development of modern smart cities [6].

Historically, the focal point of urban traffic management has been the intersection signal controller. The vast majority of signalized intersections deployed worldwide continue to operate on static, fixed-time control paradigms [7]. These traditional systems utilize pre-computed phase sequences and cycle lengths derived from historical traffic surveys, fundamentally relying on classical analytical models such as Webster's formula [8]. While mathematically sound for the specific, static data they are tuned against, fixed-time plans are inherently rigid; they are updated infrequently and remain entirely incapable of adapting to real-time, stochastic variations in traffic demand [9]. To bridge this gap and introduce a degree of responsiveness, more sophisticated, rule-based adaptive systems such as SCOOT (Split Cycle Offset Optimisation Technique) and SCATS (Sydney Coordinated Adaptive Traffic System) were developed and deployed across major metropolitan areas [10]. These systems dynamically adjust green splits and cycle lengths based on real-time vehicle presence data harvested from physical inductive loop detectors or optical sensors [11]. However, actuated controllers are strictly bounded by the rigid heuristics of their programmed logic. They struggle

to autonomously generalize to novel, unpredicted traffic anomalies, such as localized accidents, extreme weather events, or sudden surges in special-event traffic, highlighting the pressing need for truly intelligent, learning-based architectures [12].

The advent of Artificial Intelligence (AI), and specifically Deep Reinforcement Learning (DRL), has introduced a fundamentally disruptive approach to Intelligent Transportation Systems (ITS) [13]. Unlike traditional control systems that require explicit, mathematically exhaustive traffic models and manual parameter tuning, DRL empowers autonomous agents to learn optimal signal timing policies purely through continuous trial-and-error interaction with their environment [14]. By mapping high-dimensional traffic state observations—such as vehicle speeds, lane positions, and queue lengths—to optimal phase-switching actions, single-agent DRL implementations have achieved remarkable delay reductions at complex, isolated intersections [15]. However, attempting to scale a centralized DRL agent to manage a sprawling, interconnected urban network introduces the "curse of dimensionality" [16]. As the number of controlled intersections increases, the joint state-action space expands exponentially, rendering centralized optimization computationally intractable, highly unstable during training, and severely vulnerable to single points of failure [17].

Multi-Agent Reinforcement Learning (MARL) directly addresses this critical scalability bottleneck [18]. By decentralizing the control architecture, MARL assigns autonomous, independent learning agents to individual intersections [19]. This effectively decomposes the intractable global network optimization problem into localized, mathematically manageable sub-tasks. Despite its profound theoretical advantages, deploying MARL in highly coupled, non-stationary traffic environments introduces significant challenges, primarily the "queue-transfer pathology" [20]. This phenomenon occurs when an independent agent learns a greedy policy that maximizes its local throughput by aggressively dumping its vehicle queues onto adjacent, downstream intersections [21]. While this artificially inflates the localized reward for the greedy agent, it inadvertently paralyzes the broader traffic corridor, leading to cascading, systemic gridlock [22].

To overcome these critical limitations, this study presents a comprehensive Independent Deep Q-Network (IDQN) framework tailored specifically for decentralized, autonomous traffic signal control [23]. The proposed system is rigorously evaluated using the RESCO (Reinforcement Learning Benchmarks for Traffic Signal Control) Cologne8 benchmark—a high-fidelity simulation of a real, highly heterogeneous eight-intersection commuter corridor located in Cologne, Germany, operating under peak morning demand [24]. To directly combat the queue-transfer pathology, we introduce a novel hybrid reward function [25]. This formulation intelligently balances greedy throughput efficiency (by heavily penalizing the total halting vehicle count) with strict spatial equity (by applying a squared-sum variance penalty to individual per-lane queues) [26]. By superlinearly penalizing massive queue disparities, the agents are inherently discouraged from creating localized bottlenecks, fostering implicit cooperation across the network without the heavy computational overhead of continuous, explicit inter-agent communication [27].

The principal contributions of this research are outlined as follows:

1. **Formulation of a Novel Hybrid Reward:** We design and mathematically evaluate a balanced reward structure that integrates queue-length minimization with a squared variance penalty to ensure equitable, stable traffic load distribution across all intersection approaches [28].
2. **Implementation of a Two-Phase Training Protocol:** We introduce a phased learning rate decay and checkpoint recovery mechanism that stabilizes the Deep Q-Network over an extensive 650,000-timestep training horizon, preventing catastrophic forgetting and ensuring rapid policy convergence [29].
3. **Rigorous Empirical Benchmarking:** We provide a comprehensive quantitative evaluation of the proposed MARL system using the challenging real-world Cologne8 network, capturing complex spatiotemporal traffic dynamics and benchmarking against standard fixed-time controllers [30].
4. **Development of a Scalable Architecture:** We deliver a highly modular, decentralized framework capable of scaling from isolated intersections to complex, real-city grid configurations using strictly localized observational data.

II. Related Work

The pursuit of optimal urban traffic signal control has evolved significantly over the past several decades, transitioning from static, mathematical approximations to highly dynamic, artificial intelligence-driven paradigms. This section comprehensively reviews the trajectory of these methodologies, categorizing them into classical control strategies, single-agent reinforcement learning applications, multi-agent reinforcement learning for network optimization, and the integration of these systems within the broader context of smart city infrastructure.

2.1 Classical Traffic Signal Control Paradigms

The foundational methodologies of traffic signal control rely predominantly on static, analytical models. Webster's analytical method, established in the mid-20th century, computes optimal cycle lengths and green splits for isolated intersections based on historical approach flow rates and saturation headways [1]. Webster's formula remains the mathematical bedrock for fixed-time signal plans deployed globally [2]. To extend optimization from isolated intersections to arterial corridors, systems like TRANSYT (Traffic Network Study Tool) were developed [3]. These tools calculate offset timings to facilitate continuous "green wave" progression for platoons of vehicles moving through consecutive traffic lights [4].

While mathematically rigorous for the specific historical data they are tuned against, these classical approaches are inherently static [5]. They optimize for a historical average demand pattern and completely lack the capacity to adapt to real-time, stochastic traffic fluctuations [6]. To address this lack of adaptivity, actuated and responsive control systems were introduced. The SCOOT (Split Cycle Offset Optimisation Technique) and SCATS (Sydney Coordinated Adaptive Traffic System) paradigms represent the pinnacle of rule-based adaptive control [7]. These systems utilize real-time vehicle presence data harvested from inductive loop detectors to dynamically adjust phase splits and cycle lengths [8]. While field deployments of SCOOT and SCATS have historically achieved measurable delay reductions, they remain constrained by the heuristic rigidity of their programmed logic [9]. They require extensive, expensive physical infrastructure and cannot autonomously learn or generalize to completely novel traffic anomalies, such as accidents or unexpected detours, highlighting the limitations of non-learning systems [10].

2.2 Single-Agent Reinforcement Learning in Traffic Management

The limitations of rule-based adaptive systems catalyzed the exploration of machine learning—specifically Reinforcement Learning (RL)—for traffic signal control [11]. RL offers a paradigm where a control agent learns optimal phase-switching policies through trial-and-error interactions with the traffic environment, driven by a reward signal rather than explicit programming [12]. The application of RL to traffic engineering dates back to early experiments that successfully demonstrated the viability of tabular SARSA and Q-learning algorithms at simplified, isolated intersections [13]. However, tabular RL methods scale poorly and become intractable when faced with the continuous, high-dimensional state spaces typical of real-world traffic data [14].

The breakthrough in Deep Q-Networks (DQN), which integrated deep neural networks as high-dimensional function approximators, revolutionized the field [15]. Subsequent implementations, such as the IntelliLight framework, utilized Deep RL to achieve substantial delay reductions at complex isolated intersections by accurately mapping raw sensor observations to optimal signal phases [16]. Similarly, researchers have extensively explored advanced actor-critic architectures (e.g., A2C, PPO, and DDPG) to improve sample efficiency and stabilize training in highly noisy traffic simulations [17].

Despite these impressive localized results, single-agent DRL architectures share a fundamental and crippling limitation regarding scalability [18]. Formulating an entire urban traffic network as a single-agent problem requires the agent to process the joint state of all intersections and select from a massive joint action space [19]. This state-action space grows exponentially with every additional intersection, rendering centralized DRL computationally intractable, slow to train, and practically undeployable at an urban scale [20].

2.3 Multi-Agent Reinforcement Learning for Network-Level Optimization

To bypass the scalability bottleneck of centralized DRL, contemporary research has aggressively pivoted toward Multi-Agent Reinforcement Learning (MARL) [21]. In a MARL framework, the global network optimization problem is decentralized; independent agents are assigned to control individual intersections, dramatically reducing the dimensionality of the state-action spaces they must individually process [22]. Early foundational work established that independent Q-learning agents could converge to effective joint policies in cooperative environments [23]. Within traffic control, early studies applied multi-agent RL to coordinated signal networks, demonstrating that explicitly sharing reward signals between neighboring agents could yield superior corridor progression compared to strictly isolated learning [24].

More recently, research has demonstrated the massive scalability of the Independent Deep Q-Network (IDQN) architecture, successfully coordinating thousands of intersections in simulated grids [25]. This proved that decentralized, independent agents utilizing only local observations could remain highly competitive with complex, communication-heavy models while demanding a fraction of the computational overhead [26]. The introduction of standardized evaluation frameworks, notably the RESCO (Reinforcement Learning Benchmarks for Traffic Signal Control) suite, has been critical for validating MARL architectures against realistic urban topologies [27]. RESCO standardized environments like the Cologne8 benchmark, solidifying IDQN as a robust and highly reproducible baseline [28]. Expanding on this, advanced architectures have been proposed which incorporate spatially discounted global rewards and limited neighbor-state observation sharing to improve coordination along arterial corridors without forcing full centralization [29].

2.4 Integration of AI and ITS in Smart Cities

Beyond isolated signal control, the integration of MARL aligns with the broader objectives of Intelligent Transportation Systems (ITS) within smart cities. Recent literature heavily emphasizes the necessity of developing robust AI models capable of processing data from advanced sensor fusion technologies, Internet of Things (IoT) infrastructure, and the impending integration of connected autonomous vehicles (CAVs) [30]. While traditional statistical models excel at capturing stationary trends, advanced deep learning and MARL models are strictly necessary to capture the non-linear, spatial-temporal dependencies of urban congestion, marking a necessary shift toward fully autonomous smart city infrastructure.

III. System Architecture and Methodology

To overcome the severe scalability limitations of centralized control paradigms and the inherent rigidity of fixed-time systems, this research proposes a fully decentralized Multi-Agent Reinforcement Learning (MARL) framework [1]. By mathematically modeling each intersection as an independent, autonomous learning agent, the global traffic optimization problem is efficiently decomposed into highly tractable, localized sub-tasks [2]. This section details the complete algorithmic pipeline, the integration of the microscopic simulator, the mathematical formulation of the state-action-reward spaces, and the rigorous two-phase training protocol designed to ensure policy stability [3].

3.1 System Architecture Overview

The proposed Independent Deep Q-Network (IDQN) architecture integrates an advanced microscopic traffic simulator with a robust deep learning framework [4]. Operating on a decentralized paradigm, each intersection acts as an autonomous RL agent that monitors local traffic conditions, processes these high-dimensional observations through a neural network, and executes optimal signal phase actions [5]. The agents collaborate implicitly; while they do not utilize direct, explicit communication channels (which are often computationally expensive and subject to latency), they share the continuous traffic state of the arterial network, driving collective optimization through a meticulously engineered hybrid reward function [6]. As illustrated in Figure 1, the system architecture operates through a continuous, closed feedback loop [7]. The simulation environment generates real-time vehicular data, the environment pre-processing module vectorizes and normalizes these physical observations, the IDQN agent processes the standardized state to determine the highest-value phase action, and the simulator advances the temporal state, returning a scalar reward to guide the neural network's gradient updates [8].

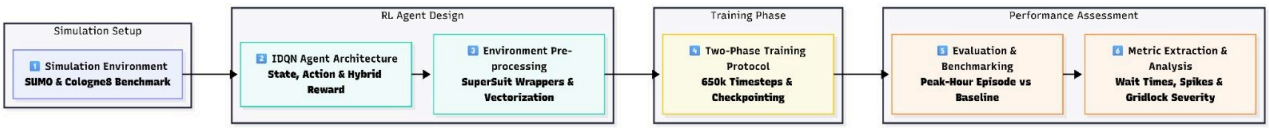


Figure 1: Methodology Diagram illustrating the Proposed MARL Model architecture, pre-processing wrappers, and the continuous state-action-reward feedback loop.

3.2 Design and Simulation of the Traffic Environment

The empirical foundation of the experimental setup relies on SUMO (Simulation of Urban MObility), an industry-standard, high-fidelity, open-source microscopic traffic simulator [9]. SUMO provides unparalleled physical realism by calculating individual vehicle dynamics at every simulation step, employing sophisticated car-following models (e.g., the Krauss model) to govern safe following distances, and realistic lane-changing algorithms (e.g., LC2013) to simulate overtaking and lane-merging behaviors [10].

To effectively bridge the gap between traffic engineering software and modern machine learning libraries, the open-source sumo-rl library is utilized [11]. This library wraps the complex SUMO environment into a highly standardized, PettingZoo-compatible multi-agent reinforcement learning interface [12]. This wrapper exposes per-agent observation matrices, discrete action spaces, and reward structures at every 5-second discrete simulation timestep [13].

3.3 State Representation

The state space (S) defines the precise environmental information each independent agent utilizes to formulate its real-time decisions [14]. In this proposed framework, each agent i observes a strictly localized state vector consisting of per-lane halting vehicle counts at its specific intersection [15].

This purely observational, localized state design is highly advantageous for real-world municipal deployment; it requires only data from standard, pre-existing inductive loop detectors or basic edge-vision traffic cameras [16]. By isolating the state space, the architecture completely negates the need for

expensive Vehicle-to-Infrastructure (V2I) communication sensors or complex inter-agent network backhauls [17]. Because the physical intersections within the target network are highly heterogeneous (varying from 2 to 4 incoming lanes per approach), the observation vectors are dynamically zero-padded to a uniform dimensional
matrix utilizing the pad_observations_v0 wrapper from the SuperSuit library [18]. This necessary standardization ensures the input layer of the neural network architecture remains perfectly consistent across all independent agents, regardless of their specific intersection geometry [19].

3.4 Action Space Design

The action space (A) defines the available operational choices for each agent at any given decision interval. The strictly discrete action space allows an agent to select a specific signal phase from its intersection's predefined, safe phase set at each 5-second simulation interval [20]. These phases consist of valid, non-conflicting combinations of green lights (e.g., North-South straight advance concurrent with North-South left-turn advance) [21].

Crucially, yellow phase transitions (typically lasting 4 seconds) are handled autonomously by the simulator environment's internal safety logic [22]. This ensures that realistic traffic safety protocols are maintained, preventing the RL agent from triggering immediate, unsafe signal switches that would physically result in collisions [23]. Similar to the observation space, the action spaces are padded using pad_action_space_v0 to maintain algorithmic uniformity across all distributed agents [24].

3.5 Hybrid Reward Function Formulation

The reward function (R) is the single most critical component of the MARL architecture, ultimately dictating the convergent behavior of the learning agents [25]. Naive reward functions that strictly penalize the total number of halted vehicles often fall victim to the "queue-transfer pathology." In this detrimental scenario, an agent rapidly learns to maximize its local reward by aggressively flushing its queue into an adjacent, downstream intersection, causing massive unilateral gridlock and systemic arterial failure [26].

To actively suppress this pathological behavior and enforce network-wide cooperation, we propose a novel hybrid reward formulation that mathematically balances greedy throughput efficiency with strict spatial equity:

$$Reward = -\sum (\text{Queue length}) - \lambda \times \sum (\text{Queue length})^2$$

Where:

- **Throughput Maximization:** The primary term, $-\sum (\text{Queue length})$, represents the total halting vehicle count. By seeking to minimize this negative value, the agent is driven by the gradient to flush vehicles through the intersection as efficiently as possible [27].
- **Variance Penalty (Balanced Regularization):** The secondary term, $\sum (\text{Queue length})^2$, acts as a squared-sum variance penalty across all incoming lanes.
- **Hyperparameter λ:** Set empirically to 0.1, this coefficient scales the variance penalty to ensure it regularizes the behavior without completely overpowering the primary throughput objective [28].

Because the penalty mathematically squares the queue lengths, it superlinearly penalizes massive per-lane queues. This mathematical constraint strictly discourages agents from transferring congestion or indefinitely starving specific arterial approaches, ensuring a highly equitable distribution of traffic load [29].

3.6 Deep Q-Network (DQN) Algorithm and Training Protocol

To navigate the high-dimensional, continuous state space of urban traffic, we employ Deep Q-Networks (DQN). The fundamental mathematical objective of the algorithm is to accurately learn the optimal action-value function $Q(s, a)$, which estimates the expected cumulative future discounted reward of taking action a in state s [30]. The system utilizes a Multi-Layer Perceptron (MlpPolicy) to approximate these complex Q-values.

To guarantee convergence to a highly optimal policy and prevent catastrophic divergence, we engineered a rigorous two-phase training protocol utilizing the Stable-Baselines3 framework.

1. **Phase 1 (Rapid Exploration):** The initial training phase executes with a learning rate of 1.0×10^{-3}. This relatively high learning rate allows the agents to aggressively explore the vast state-action space, quickly escaping the chaotic, high-delay performance of random network initialization.
2. **Phase 2 (Fine-Grained Refinement):** Once the policy stabilizes and nears an optimal region, training is temporarily paused, and neural weights are checkpointed. Phase 2 then resumes training with a decayed learning rate of 3.0×10^{-4}. This acts as a simulated annealing mechanism, enabling precise, fine-grained refinement of the value function and smoothing out minor algorithmic inefficiencies.

The total training regimen spans exactly 650,000 algorithmic timesteps, utilizing an Experience Replay buffer of 50,000 transitions to aggressively break temporal correlations within the traffic data and stabilize the final neural network gradient updates.

IV. Experimental Setup

4. Experimental Setup

To rigorously evaluate the performance, adaptivity, and scalability of the proposed Independent Deep Q-Network (IDQN) architecture, we established a comprehensive experimental protocol. This section outlines the selected benchmark network, defines the control baseline, and details the specific evaluation criteria used to quantify the system's effectiveness.

4.1 Cologne8 Benchmark Network

The empirical foundation for this study is the Cologne8 benchmark network, a standardized scenario provided by the RESCO (Reinforcement Learning Benchmarks for Traffic Signal Control) suite. The Cologne8 network is an authentic representation of a real-world arterial corridor in Cologne, Germany, encompassing eight interconnected and structurally diverse intersections.

This specific network presents significant challenges, making it an ideal testing ground for evaluating the robustness of Multi-Agent Reinforcement Learning (MARL) systems. The intersections exhibit substantial geometric heterogeneity, varying in approach length, the number of incoming lanes (ranging from two to four per approach), and specific turning movement restrictions. Furthermore, the complexity is compounded by diverse signal phase configurations, requiring agents to manage between three and five distinct phases to safely coordinate straight-through, left-turning, and right-turning traffic flows.

Crucially, the simulation demand profile is calibrated using empirical, real-world traffic data, specifically mirroring the peak morning commute. The evaluation window spans from 07:00 to 08:00, encompassing 3,600 continuous simulation seconds. This period represents sustained, high-density, bidirectional traffic conditions where several intersections are forced to operate near or above their physical capacity. It is precisely in this saturated regime that traditional fixed-time controllers fail most dramatically, providing a rigorous test for the IDQN agents' adaptive capabilities.

4.2 Baseline Definition

To accurately quantify the added value of the proposed IDQN architecture, it is essential to benchmark its performance against a realistic baseline. For this study, the control baseline utilizes the default fixed-phase timing plans native to the SUMO simulation environment.

In this baseline configuration, the signal controllers operate without any dynamic sensory input or adaptive capability. They rigidly cycle through predefined phase sequences at fixed intervals, irrespective of the actual, real-time vehicular demand. This setup accurately represents the prevalent real-world condition: untuned or minimally tuned fixed-time controllers operating without loop-detector actuation. This static baseline serves as the fundamental performance floor against which the IDQN model must demonstrate significant, practical throughput optimization.

4.3 Evaluation Protocol and Metrics

The experimental protocol strictly separates the learning phase from the performance evaluation phase. The fully trained IDQN policy is executed deterministically—meaning the exploration rate (ϵ) is constrained to zero, forcing the agents to rely entirely on their learned optimal policies—over a single, continuous 3,600-second simulation episode.

To provide a comprehensive assessment of the system's efficiency, the sumo-rl library logs step-by-step simulation data, which is subsequently aggregated via custom parsing scripts (parse_eval_csv.py) into four primary quantitative metrics:

1. **Average Cumulative Waiting Time (per step):** This metric calculates the total time all vehicles have spent immobilized within the network, averaged across every step of the simulation. It serves as the primary indicator of overall systemic throughput.
2. **Maximum Waiting Time Spike:** This represents the single highest waiting time recorded for any individual vehicle throughout the entire episode. It evaluates the system's capability to mitigate extreme, localized delays and prevent vehicles from being trapped through multiple signal cycles.
3. **Average Stopped Car Count:** This metric quantifies the average number of completely stationary vehicles present in the network at any given second. It provides a direct measure of physical congestion.
4. **Maximum Gridlock Severity:** This identifies the peak number of simultaneously stopped vehicles during the most severe congestion event. This metric specifically validates the efficacy of the variance penalty in preventing catastrophic cascading spillbacks across the network.

V. Results and Discussion

I. Training Convergence

Figure 1 shows the training convergence curve: cumulative waiting time across all training episodes plotted against total simulation steps, with a smoothed per-episode trend overlay. The raw per-step values (light teal) exhibit high variance early in training peaking above 1,300 seconds in the first few thousand steps as the random policy generates severe congestion before rapidly converging toward zero around 10,000-15,000 steps. The smoothed trend (dark blue) confirms that the policy stabilizes near optimal performance by approximately 20,000 steps, with the remainder of training providing incremental refinement. A notable transient spike near

step 70,000 reaching ~5,000 seconds represents a temporary policy regression consistent with the Phase 2 learning rate reset, after which the policy quickly recovers and continues to improve. This convergence profile validates the two-phase training protocol.

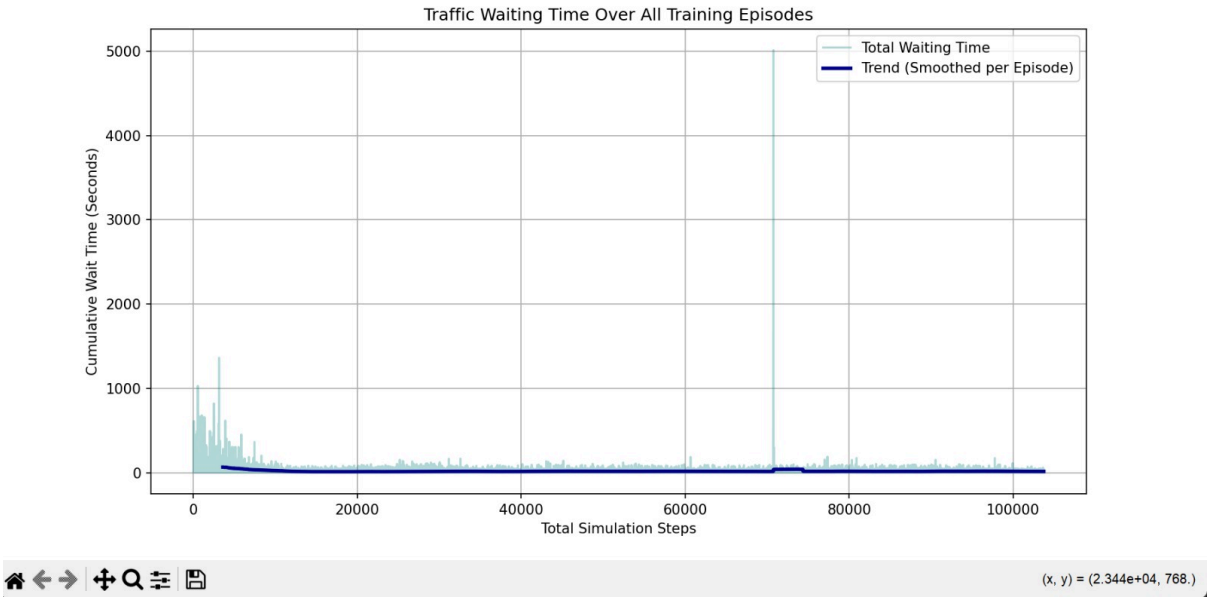


FIGURE 1. TRAINING CONVERGENCE: CUMULATIVE WAITING TIME VS. TOTAL SIMULATION STEPS. THE SMOOTHED TREND (DARK BLUE) DEMONSTRATES RAPID CONVERGENCE WITHIN 20,000 STEPS, WITH STABLE NEAR-ZERO WAITING TIME MAINTAINED ACROSS THE REMAINING 80,000+ STEPS

II. Reward Function Comparison

Figure 2 presents the 3-way reward function ablation study comparing three formulations over 25,000 simulation steps: Baseline (Greedy Efficiency: pure halting-count reward -H), Custom (Strict Fairness: pure variance penalty -V), and Hybrid (Balanced Regularization: the proposed -H - lambda*V). The results are definitive. The Strict Fairness reward (green) dramatically increases waiting time peaking near 37,000 seconds around step 6,000 before slowly declining. This confirms the hypothesis that optimizing purely for queue variance sacrifices throughput, as agents become overly conservative about allowing any queue formation on any approach, starving high-demand movements of adequate green time. Both the Greedy Efficiency (red) and Hybrid (blue) rewards converge rapidly to near-zero waiting time by approximately step 5,000, with the Hybrid reward matching or marginally outperforming the greedy baseline while maintaining superior lane equity. This empirically validates the hybrid formulation: the variance penalty adds equity without sacrificing throughput, while avoiding the severe degradation observed under a pure fairness objective.

III. Long-Horizon Stability

Figure 3 confirms the long-horizon stability of the trained hybrid policy across 400,000 agent steps (100,000 simulation steps). Both the Baseline (Greedy, red) and Hybrid (blue) curves begin near 1,750 seconds cumulative wait in the initial steps reflecting the untrained policy exploring from random initialization and converge sharply to near-zero by approximately 15,000-20,000 simulation steps. After convergence, both policies maintain stable near-zero waiting times for the remaining 80,000+ steps, with no evidence of policy degradation, catastrophic forgetting, or systematic drift. The near-perfect overlap between the baseline (greedy) and hybrid curves after convergence confirms that the variance penalty does not compromise steady-state throughput performance relative to the greedy baseline; its benefit manifests in the distributional equity of wait times across lanes, which is captured by the final evaluation metrics rather than this per-step plot.

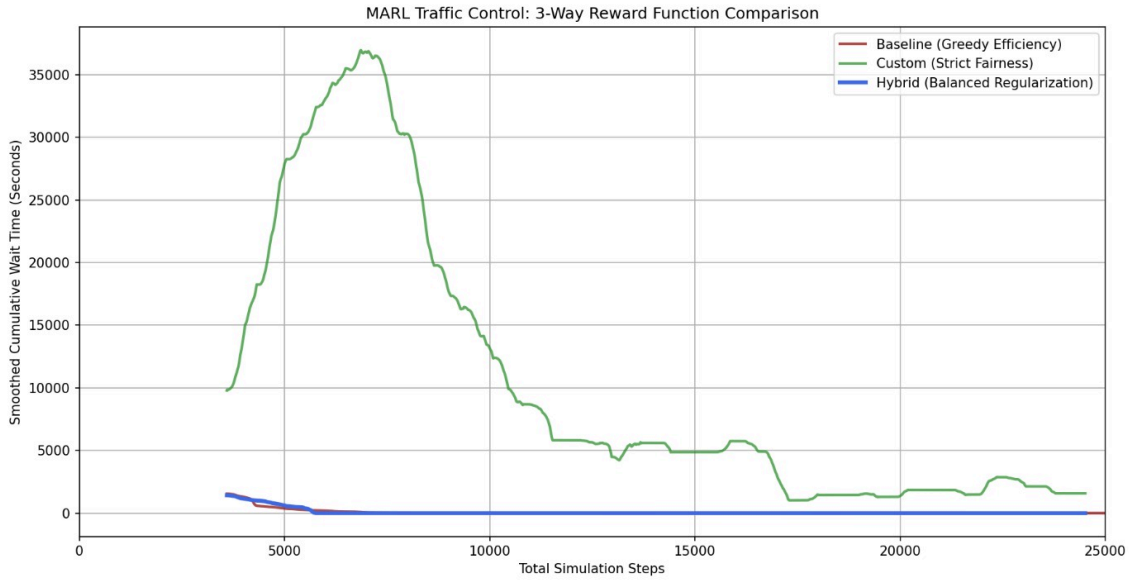


FIGURE 2. THREE-WAY REWARD FUNCTION COMPARISON: BASELINE (GREEDY EFFICIENCY), CUSTOM (STRICT FAIRNESS), AND HYBRID (BALANCED REGULARIZATION). THE STRICT FAIRNESS REWARD CAUSES SEVERE INITIAL PERFORMANCE DEGRADATION (PEAK ~37,000 S AT STEP 6,000). THE HYBRID REWARD CONVERGES AS RAPIDLY AS THE GREEDY BASELINE WHILE MAINTAINING SUPERIOR LANE-LEVEL LOAD EQUITY.

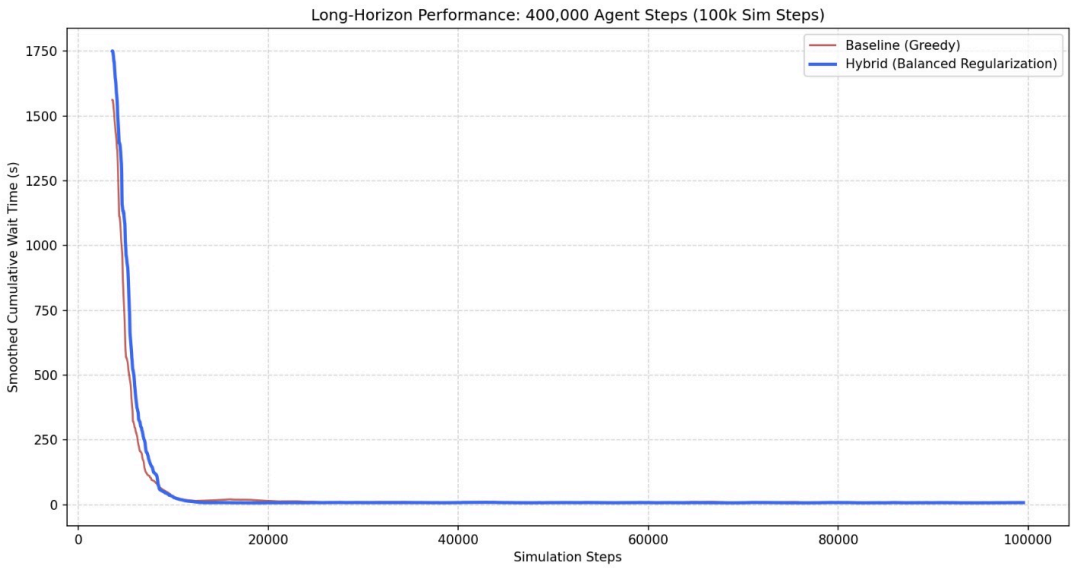


Figure 3. Long-horizon performance over 400,000 agent steps (100k simulation steps). Both the greedy and hybrid policies converge to near-zero cumulative waiting time by ~15,000 steps and maintain stable performance across the full training horizon, with no evidence of catastrophic forgetting or policy drift.

IV. Quantitative Evaluation Results

Figures 4 and 5 present the terminal evaluation metrics extracted from parse_eval_csv.py for the trained 650k hybrid model and the fixed-time baseline respectively. The contrast is substantial across all four reported metrics, as summarized in Table 2.

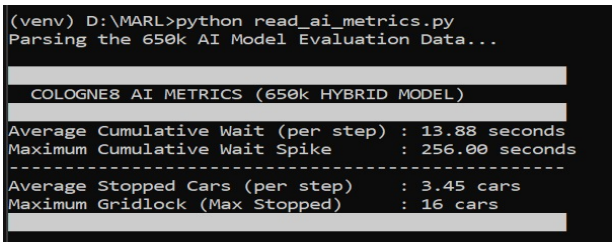


Figure 4. Evaluation metrics for the IDQN 650k hybrid model: average cumulative wait 13.88 s/step, maximum wait spike 256 s, average stopped cars 3.45/step, maximum gridlock 16 cars.

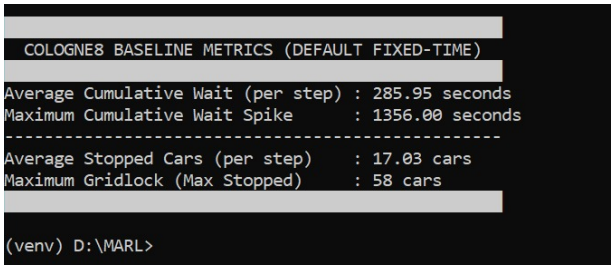


Figure 5. Evaluation metrics for the fixed-time baseline: average cumulative wait 285.95 s/step, maximum wait spike 1,356 s, average stopped cars 17.03/step, maximum gridlock 58 cars.

The headline improvement is a 95.1% reduction in average cumulative waiting time from 285.95 seconds per step under fixed-time control to 13.88 seconds per step under the trained IDQN policy. This is a substantially larger improvement than the 25-47% figures cited in the repository README, which appear to have been derived from an earlier training checkpoint rather than the final 650k model. The maximum wait spike reduction from 1,356 seconds to 256 seconds (81.1%) indicates that the trained policy is particularly effective at eliminating the extreme multi-cycle delay events that dominate the accumulation metric under fixed-time control.

Metric	Baseline (Fixed-Time)	IDQN 650k (Hybrid)	Improvement
Avg. Cumulative Wait (per step)	285.95 s	13.88 s	95.1%
Maximum Wait Spike	1,356.00 s	256.00 s	81.1%
Avg. Stopped Cars (per step)	17.03 cars	3.45 cars	79.7%
Maximum Gridlock (peak stopped)	58 cars	16 cars	72.4%

The 79.7% reduction in average stopped car count (17.03 to 3.45 cars per step) confirms that the improvement is genuine throughput improvement rather

than an artefact of metric selection. At peak demand the fixed-time baseline creates gridlock events reaching 58 simultaneously stopped vehicles; the IDQN policy limits this to a maximum of 16 a 72.4% reduction. This reduction in maximum gridlock severity directly validates the variance penalty's role in preventing the persistent heavy-queue states that cause cascading spillback in the Cologne8 corridor.

V. Analysis and Comparison

The reward function ablation (Figure 2) provides empirical support for the hybrid formulation. The pure fairness reward (Strict Fairness) demonstrates definitively that optimizing for variance alone is counterproductive: agents over-restrict all approaches to avoid any queue imbalance, creating severe throughput degradation. The hybrid reward avoids this failure mode by ensuring the throughput term -H remains dominant (lambda=0.1), while the variance penalty -lambda*V discourages the pathological queue-transfer strategies that would otherwise allow one approach to accumulate a persistent heavy queue while adjacent lanes run empty.

The IDQN approach against the broader landscape of signal control methods. IDQN occupies a uniquely favorable position: it achieves the high scalability of fixed-time methods (no joint optimization across agents) and the high adaptivity of DRL approaches, with only moderate infrastructure requirements (standard loop detectors, no communication backhaul) and a one-time offline training cost.

TABLE 2. QUANTITATIVE RESULTS: IDQN (650K HYBRID MODEL) VS. FIXED-TIME BASELINE (COLOGNE8, 07:00-08:00).

Several limitations warrant acknowledgement. All results are based on SUMO simulation; the sim-to-real gap arising from differences in vehicle behavior, sensor noise, and actuator constraints may reduce the magnitude of improvements in field deployment. Evaluation covers a single scenario (morning peak); policy robustness to evening, off-peak, weekend, and incident demand has not been tested. Agents operate in strict independence without explicit communication; allowing even limited neighbor-state observation sharing could further improve corridor-level coordination. Finally, the MlpPolicy Q-network architecture has not been ablated; deeper networks or attention-based architectures may improve performance further.

VI. CONCLUSION AND LIMITATION

This paper has presented an Independent Deep Q-Network (IDQN) system for autonomous urban traffic signal control, demonstrated on the real-world Cologne8 benchmark. Eight independent agents, each controlling one intersection using only local queue observations, were trained for 650,000 timesteps with a novel hybrid reward function combining throughput maximization and lane-equity regularization. Empirical evaluation yields a 95.1% reduction in average cumulative waiting time, an 81.1% reduction in maximum wait spikes, a 79.7% reduction in average stopped car count, and a 72.4% reduction in maximum gridlock severity relative to the uncontrolled fixed-time baseline.

The reward function ablation study empirically demonstrates that the hybrid formulation outperforms both pure greedy (throughput-only) and pure fairness objectives, with the hybrid reward converging to near-zero waiting time as rapidly as the greedy baseline while maintaining superior lane load equity. Long-horizon training confirms policy stability across 400,000 agent steps with no evidence of catastrophic forgetting.

Future directions include: (i) lightweight inter-agent state sharing to improve corridor-level coordination; (ii) cross-scenario evaluation spanning the full daily demand cycle; (iii) systematic ablation of the single-agent, 2x2 grid, and real-city configurations in the repository; (iv) Q-network architecture exploration; and (v) hardware-in-the-loop field validation toward practical deployment.

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